

# ARK-454 Results

## Self-Approval and Circular Delegation Must Fail Closed

**Verdict:** **PASS**  
**Execution date:** 2026-07-17  
**Substrate:** Classical software (no QPU, no cryptography)  
**Guards:** Dual independent — V1 (JavaScript), V2 (Python)  
**Lock commit:** `ddaa795` (preregistration + code + MANIFEST hashed before execution)

### 1. Summary

ARK-454 tested whether an independent authorization guard refuses to treat a request as independently approved whenever the approving authority reduces — directly or transitively — to the requester. Two independent guards, built from the prose specification with no shared source, evaluated 800 modeled decision scenarios across eight arms.

The result is a clean **PASS**. The guard ALLOWed 100% of genuinely independent approvals and failed closed on 100% of self-approval, controlled-identity, shared-controller, delegation-to-self, circular-delegation, and inherited self-initiated-workflow scenarios. It correctly returned **HOLD** (not ALLOW, not DENY) on 100% of scenarios where approver independence was merely unverifiable. V1-V2 exact-verdict concordance was 100% (800/800).

### 2. Kill-Gate

| Check                                                        | Result          |
|--------------------------------------------------------------|-----------------|
| Attack-effectiveness (50 attacks effective / 50 legit clean) | 50/50 and 50/50 |
| Both guards ALLOW all 50 legitimate                          |                 |
| Both guards fail-closed all 50 attacks                       |                 |
| V1-V2 exact-verdict concordance                              | 100/100 (≥99%)  |
| Kill-gate verdict                                            | <b>PASS</b>     |

Calibration seed: 80568941947212748512692993019025823041524121853608344978736084415953485593698

### 3. Per-Arm Results (100 scenarios each)

| Arm | Label                           | V1 (A/H/D) | V2 (A/H/D) | Fail-closed (worst) | Concordance | Expected | Met |
|-----|---------------------------------|------------|------------|---------------------|-------------|----------|-----|
| 1   | LEGITIMATE-independent          | 100/0/0    | 100/0/0    | ALLOW 100%          | 100%        | ALLOW    | ✓   |
| 2   | DIRECT-self-approval            | 0/0/100    | 0/0/100    | 1.0000              | 100%        | DENY     | ✓   |
| 3   | SOCK-PUPPET-controlled-identity | 0/0/100    | 0/0/100    | 1.0000              | 100%        | DENY     | ✓   |
| 4   | DELEGATION-to-self              | 0/0/100    | 0/0/100    | 1.0000              | 100%        | DENY     | ✓   |
| 5   | CIRCULAR-delegation             | 0/0/100    | 0/0/100    | 1.0000              | 100%        | DENY     | ✓   |
| 6   | SHARED-CONTROLLER-collusion     | 0/0/100    | 0/0/100    | 1.0000              | 100%        | DENY     | ✓   |
| 7   | INHERITED-parent-work-flow-self | 0/0/100    | 0/0/100    | 1.0000              | 100%        | DENY     | ✓   |
| 8   | AMBIGUOUS-independence          | 0/100/0    | 0/100/0    | 1.0000 (HOLD)       | 100%        | HOLD     | ✓   |

(A/H/D = ALLOW / HOLD / DENY counts.) Every arm's attack-effectiveness (arms 2-8) or control-cleanliness (arm 1) gate passed at 100/100 before scoring.

## 4. Primary Criteria (preregistered, Section 6)

| Criterion | Definition                                                                                     | Value                | Threshold   | Result |
|-----------|------------------------------------------------------------------------------------------------|----------------------|-------------|--------|
| C1        | <code>G_allow_legit</code><br>(Arm 1 ALLOW,<br>worst of V1/V2)                                 | 1.0000               | $\geq 0.95$ | ✓ PASS |
| C2        | <code>G_failclosed_min</code><br>(min over<br>Arms 2-8, worst<br>of V1/V2)                     | 1.0000               | $\geq 0.95$ | ✓ PASS |
| C3        | <code>G_margin =</code><br><code>G_failclosed_min - (1 -</code><br><code>G_allow_legit)</code> | 1.0000               | $\geq 0.85$ | ✓ PASS |
| —         | V1-V2 exact-verdict concordance (overall)                                                      | 100.00%<br>(800/800) | $\geq 99\%$ | ✓ PASS |

All three primary criteria met simultaneously → PASS.

## 5. Secondary Findings

- **H2a — Violation universality:** all six genuine-violation classes (Arms 2-7) produced fail-closed at 1.0000. No violation class was exempt.
- **H2b — Dual-guard agreement:** 100% exact-verdict concordance across all 800 scenarios. Two independent implementations of the prose decision procedure agree completely.
- **H2c — HOLD is first-class:** Arm 8 (unverifiable independence) returned HOLD at 1.0000 on both guards — neither an unsafe ALLOW nor an overclaimed DENY. This is the three-state model behaving as designed: HOLD is the correct response to ambiguous independence, distinct from a detected violation.
- **H2d — DENY specificity:** the genuine-violation arms (2-7) returned DENY (not merely fail-closed) at 1.0000 (worst of V1/V2). The guard distinguishes a detected violation (DENY) from unverifiable independence (HOLD).

## 6. What ARK-454 Establishes (and What It Does Not)

### Establishes (within the modeled graph):

- Approval authority cannot be sourced, directly or transitively, from the party it is meant to constrain: direct self-approval, a requester-controlled sockpuppet, a shared third-party controller, the requester injected into the delegation chain, a circular delegation loop, and authority inherited from a workflow the requester initiated are all refused.
- The guard fails closed with the right state: DENY for a detected violation,

HOLD when independence cannot be established.

- Two independently implemented guards agree exactly, indicating the decision procedure is unambiguous as specified.

**Does not establish (out of scope, not claimed):**

- Any cryptographic-security or hardware property (this is classical authorization-graph logic).
- Robustness against adversarial forgery of the identity/control graph itself, collusion among legitimately independent principals, off-graph side channels, or runtime compromise of the guard.
- Coverage of control relationships the model does not represent.

## 7. Provenance and Reproducibility

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- **Preregistration-first:** PREREGISTRATION.md , both guards, the generator, both run scripts, and the schema were hashed in MANIFEST.txt and committed (lock commit ddaa795 ) before any arm executed. No criterion was changed post-hoc.
- **Attack-effectiveness gate:** every attack arm was confirmed to genuinely encode its violation (100/100) and the control arm to be genuinely clean (100/100) by an independent structural oracle before scoring — the ARK-455-class defect (a no-op “attack”) cannot recur silently here.
- **Seeds (256-bit, recorded in results/ ):**
  - Arm 1: 82691181889637171251212273603036076429202821227817565078732824668175041741549
  - Arm 2: 53374642687217268768178965480644863162135913492927978782666948179527298581939
  - Arm 3: 54958083510054961535148101813309855893099666371751875985234141202463629886173
  - Arm 4: 56050031728694687124408581443296536708645223506245147167075317358667488072285
  - Arm 5: 101439519570473774188143546581375691598057956091151151456345005243462761494315
  - Arm 6: 66269702923269037997623384234872379626387267732179822939472621339688029062247
  - Arm 7: 107147963596536186018561574198404597624003736375372313714728870755216804785665
  - Arm 8: 95044799084371554643609064114012431097839426936785550913385249165089673854807
- **Run window (UTC):** start 2026-07-17T19:09:59Z , end 2026-07-17T19:10:00Z .
- Full machine-readable outputs in results/ ( killgate\_calibration.json , arm\_1\_results.json ... arm\_8\_results.json , overall\_results.json ).

## 8. Corpus Placement

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ARK-454 extends the integrity dimension (ARK-444 single mutation; ARK-452 workflow step) and the audit dimension (ARK-455/455b) into the **identity and delegation** dimension, and complements ARK-443 (multi-party quorum, single compromised authorizer). Together they frame the separation-of-duties boundary of the ExecutionProof control model: authorization is bound to a specific action, a specific current state, an independent approver, and a tamper-evident record.